Create an AWS EC2 Web App with Load Balancing

Step 1
Launch two EC2 instances using the Amazon Linux 2 AMI and the t2.micro instance type.

Step 2
Create and configure a security group that allows inbound traffic on port 22 (SSH) and port 80 (HTTP).

Step 3
Note the public IP address or public DNS of each instance for later testing.

Step 4
Connect to each EC2 instance via SSH using your terminal or preferred SSH client.

Step 5
Deploy a basic HTML page for each instance using a simple web server such as a nginx.

```
sudo dnf update -y

sudo dnf install nginx -y

sudo systemctl start nginx

sudo systemctl enable nginx

sudo systemctl status nginx

cat /usr/share/nginx/html/index.html

sudo vim  /usr/share/nginx/html/index.html
```

Add

```
<!DOCTYPE html>
<html>
<head>
   <title>30-Days-AWS</title>
   <meta charset="UTF-8">
</head>
<body>
   <h1>Welcome to 30-Days-AWS</h1>
</body>
</html>
```

```
cat /usr/share/nginx/html/index.html
```

Step 6
Open the public IP of each instance in your browser to confirm that the web page loads successfully.

http://public-ip/

Step 7
Navigate to the EC2 Console and create an Application Load Balancer with an internet-facing scheme using HTTP on port 80.

Step 8
Create a target group with instance as the target type and add both EC2 instances to it.

Step 9
Configure the health check path and associate the target group with the load balancer before completing the setup.

Step 10
Access the DNS name of the load balancer in your web browser.

Step 11
Refresh the page several times and observe alternating responses from each instance to confirm load distribution.

Step 12
Optionally, stop one instance to test whether the load balancer routes all traffic to the healthy instance, demonstrating basic fault tolerance.